

AWR 4-Channel RF/IF Simulation Guide

Practical setup for isolation, gain matching, phase matching, VSWR, and switching-time simulations.

1. Channel-to-Channel Isolation

For a 4-port network, channel isolation is obtained from the appropriate S-parameter transmission terms.

For example, CH1→CH2 is S21, CH1→CH3 is S31, and CH1→CH4 is S41.

The reverse directions are S12, S13, and S14. For a reciprocal passive network, the unique pairings may be sufficient; for an active or non-reciprocal system, measure both directions.

2. LO→RF and LO→IF Isolation

For mixer/system-level LO leakage, use Harmonic Balance rather than an ordinary linear S-parameter analysis.

Turn on the LO and keep the RF input inactive. Observe the LO-frequency component at the RF and IF outputs.

Example: with LO = 16 GHz and LO power = +8 dBm, if the LO-frequency leakage at an output is –55 dBm, the isolation is 63 dB.

Mixer isolation is measured at the mixer RF/IF ports; system-level isolation is measured at the final outputs and therefore includes filters/amplifiers.

3. RF and IF Hierarchical Blocks

Create separate hierarchical blocks such as RF_BLOCK and IF_BLOCK. Put the internal RF components inside RF_BLOCK and the internal IF components inside IF_BLOCK.

Then create a top-level COMPLETE_SYSTEM schematic containing the RF block, mixer/LO interface as required, and IF block.

Typical signal flow is: RF INPUT → RF_BLOCK → MIXER → IF_BLOCK → IF OUTPUT. Keep the LO/mixer visible at the top level initially if you need mixer isolation measurements.

4. Gain Matching — ±1.5 dB

Run an S-parameter simulation across the operating band. Plot the transmission gain for all four channels.

Example port mapping, if CH1–CH4 use ports 1–4 as inputs and 5–8 as outputs:

CH1 = S51, CH2 = S62, CH3 = S73, CH4 = S84.

These port numbers are examples only; use your actual schematic ports.

For channel matching, calculate each channel's deviation from the average:

$$G_{avg} = (G1 + G2 + G3 + G4) / 4$$

$$Error_n = G_n - G_{avg}$$

The matching requirement is $-1.5 \text{ dB} \leq Error_n \leq +1.5 \text{ dB}$.

Create a rectangular graph with all four gains and, if required, ±1.5 dB limit lines.

5. Phase Matching — ±10°

Plot the phase of each channel transmission response, for example angle(S51), angle(S62), angle(S73), and angle(S84).

For reference-based matching, use CH1 as the reference:

$$\Delta\phi_2 = \phi_2 - \phi_1, \Delta\phi_3 = \phi_3 - \phi_1, \Delta\phi_4 = \phi_4 - \phi_1.$$

Add +10° and –10° horizontal limits.

If the specification means maximum phase difference between any two channels must be ≤10°, use the

all-channel spread:

$$\max(\phi_1, \phi_2, \phi_3, \phi_4) - \min(\phi_1, \phi_2, \phi_3, \phi_4) \leq 10^\circ.$$

Be careful about phase wrapping around $\pm 180^\circ$; use unwrapped phase where appropriate.

6. Input and Output VSWR — 1.6:1

For a linear RF/IF channel, input VSWR comes from the input reflection coefficient and output VSWR from the output reflection coefficient.

$$\text{VSWR} = (1 + |S|) / (1 - |S|).$$

For a simple 2-port channel, input VSWR uses S11 and output VSWR uses S22.

For a four-channel example with CH1 ports 1→5, CH2 ports 2→6, CH3 ports 3→7, CH4 ports 4→8:

Input VSWR: S11, S22, S33, S44.

Output VSWR: S55, S66, S77, S88.

These are example mappings only.

Create separate graphs named 4CH_INPUT_VSWR and 4CH_OUTPUT_VSWR and add a 1.6:1 limit line.

7. Switching Time — <100 ns

Switching time is a time-domain/transient requirement. Apply a time-varying control pulse to the switch control input and run a transient simulation.

A practical starting point is a simulation window of about 0–500 ns with a time step around 0.1–1 ns, adjusted for model accuracy and convergence.

Measure the RF/IF output envelope or power rather than the instantaneous RF sinusoid.

Using a 10%→90% definition:

$$T_{\text{switch}} = T_{90} - T_{10}.$$

Check both ON→OFF and OFF→ON transitions and use the worst channel/transition for the requirement.

Example: if the four channels give 72 ns, 81 ns, 68 ns, and 93 ns, the worst case is 93 ns, which satisfies a <100 ns requirement.

8. Recommended AWR Report Structure

Create the following report graphs:

- 1) 4CH_GAIN — four channel transmission gains.
- 2) 4CH_GAIN_ERROR — channel deviations from the average, with ± 1.5 dB limits.
- 3) 4CH_PHASE — four channel phases.
- 4) PHASE_MATCHING — phase errors or all-channel phase spread, with $\pm 10^\circ$ limits as applicable.
- 5) 4CH_INPUT_VSWR — four input VSWR curves with 1.6 limit.
- 6) 4CH_OUTPUT_VSWR — four output VSWR curves with 1.6 limit.
- 7) LO_RF_ISOLATION — LO leakage at RF output.
- 8) LO_IF_ISOLATION — LO leakage at IF output.
- 9) SWITCHING_TIME — transient/envelope response and measured transition time.

Use the AWR equation/measurement editor to insert the exact measurement names rather than guessing syntax, because the available syntax can depend on the AWR version and analysis setup.